HOW DO WE KNOW WHAT WE KNOW?

Evidence, falsification, reproducibility, and real-world application of an adversarial anomaly-detection framework

THE STANDARD

A claim earns confidence only if it survives algebra, simulation, adversarial critique, and independent reproduction.

Independent research • v1.4.2 protected benchmark closed

EPISTEMOLOGY

First: separate the claims

The mathematics may be valid even if the motivating thought experiment is false.

CLAIM A

The speculative population exists.

Status: unsupported by this work.

CLAIM B

The detection mathematics is internally correct.

Status: strongly checked within the stated models.

CLAIM C

The framework transfers to real sensor systems.

Status: testable next; not yet established.

PLAIN ENGLISH

This deck is mostly about Claim B and how to move honestly toward Claim C. It does not use mathematical elegance as evidence for Claim A.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

EVIDENCE

Our evidence ladder

Different claims deserve different kinds of confidence.

1. Exact identities	Algebraic equalities and model reductions
2. Analytic predictions	Closed forms, χ^2/noncentral-χ^2 power
3. Seeded simulation	Direct draws from the stated generative model
4. Regression to archived results	675 historical Monte Carlo cells
5. Cross-machine regeneration	Different NumPy/LAPACK builds reproduce outputs

PLAIN ENGLISH

The ladder is deliberately cumulative: simulation does not replace algebra, and reproducibility does not prove the model describes nature. Each rung answers a different failure mode.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

01

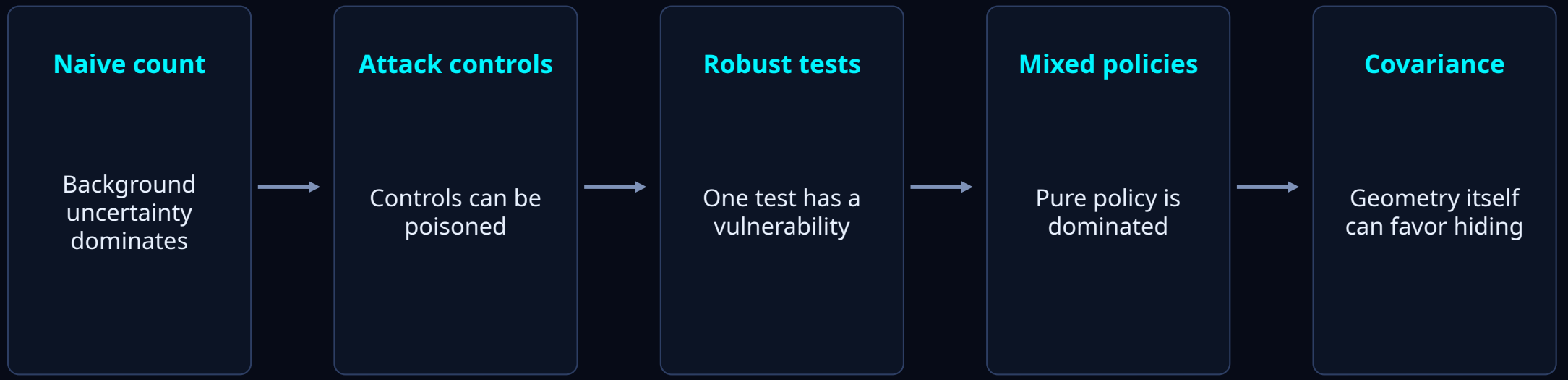
What survived repeated attempts to break it

The strongest results are the ones that remained after assumptions, utilities, estimators, grids, and randomization were changed.

The research progressed by breaking its own claims

FALSIFICATION

Several “results” disappeared when we changed an assumption. That is evidence of a healthy method.



PLAIN ENGLISH

We trust the current framework more because it contains the scars of failed simplifications: background uncertainty, contamination cost, pure-policy commitment, and covariance misspecification all changed earlier conclusions.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

IDENTIFIABILITY

Result 1 — repeated observation does not fix non-identifiability

A perfect estimate of the wrong aggregate can still leave the scientific question unanswered.

$$Y_t \sim \text{Poisson}(B + s)$$

$$\mathbb{E}[Y_t] = B + s$$

Even as T → ∞, the observations identify B+s, not B and s separately.

WHY WE BELIEVE IT

This is structural, not Monte Carlo. Two parameter pairs with the same sum generate the same observation law. No amount of repetition can separate them without an additional contrast or constraint.

PLAIN ENGLISH

More data is not automatically more information. You need a measurement that changes what is identifiable, not just another copy of the same measurement.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

Result 2 — an adaptive hider chooses its detectability

CLOSED FORM

A closed-form toy model showed exactly where the cost assumption enters.

$$D_{\mathrm{KL}} \approx \frac{s^2}{2V}, \qquad J(s) = \frac{s^2}{2V} + \kappa \log\left(\frac{100}{s}\right)$$

$$s^* = \sqrt{\kappa V}, \qquad D_{\mathrm{KL}}^* = \frac{\kappa}{2}$$

WHY WE BELIEVE IT

Differentiate J(s), set the derivative to zero, and verify the second derivative is positive. The result is exact for this objective once the quadratic KL approximation is accepted.

PLAIN ENGLISH

This did not prove that real adversaries use a logarithmic cost. It proved something more useful: the operating-cost model can dominate the equilibrium, so sensitivity analysis is mandatory.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

Result 3 — private randomization only helps under reach limits

COMBINATORICS

Coverage is an exact combinatorial statement, not a simulation artifact.

P(all sampled controls contaminated)

= C(r,k) / C(m,k), when k ≤ r

= 0, when k > r

WHY WE BELIEVE IT

Count the k-subsets entirely contained inside the r reachable controls, then divide by all k-subsets of the m candidates. If k exceeds r, such a subset cannot exist.

50%	21.4%	1.43%	0%
m=8,r=4,k=1	k=2	k=4	k=5

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

Result 4 — the background estimator is part of the attack surface

COUNTEREXAMPLE

The v1.2 spread attack falsified a seemingly reasonable detector.

ATTACK

$$x_i \mapsto x_i + \delta_i, \qquad \sum_i \delta_i \approx 0$$

Balanced +δ/−δ offsets can keep the control mean stable while inflating empirical scale.

OBSERVED FAILURE

Example one-sided detector:

0.774	0.007	≈0
k=4, δ=0	k=4, δ=10	k=4, δ=20

The detector lost power because it trusted the attacked controls to estimate its own uncertainty.

PLAIN ENGLISH

This is exactly the kind of result we value most: a concrete attack that makes a nominally sensible method fail for a reason we can explain.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

Result 5 — randomized test mixtures improve the frontier

GAME THEORY

The observer can commit to a distribution of tests so the hider cannot tailor one attack to one known rule.



WHAT THE CURVES MEAN

- Lower is better for the observer: the hider gets less utility from plain activity.
- Moving right permits more contamination advantage.
- Larger τ means more heterogeneous background, which weakens the observer.
- The optimum often mixes omnibus and combined tests rather than choosing one.

Source: mixed_policy_frontier_v1_3_5.csv, exact finite-grid LP.

02

Why covariance changed the problem

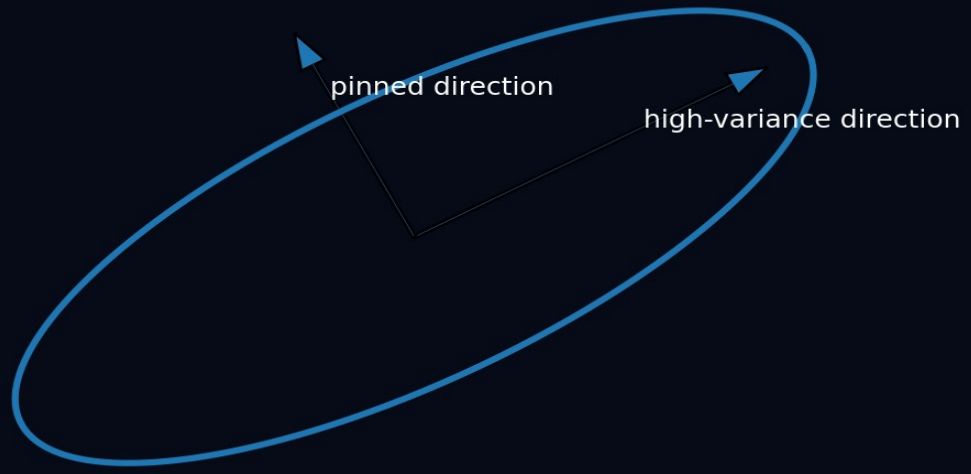
Once channels are correlated, “more noise” is not one number. The direction of the footprint relative to Σ determines detectability.

COVARIANCE

Result 6 — detectability is geometric

A footprint can be easy or hard to detect depending on which covariance direction it occupies.

Σ defines where ordinary background leaves room to hide



$$Q = z^T \Omega^{-1} z$$

$$\lambda = \mu^T \Omega^{-1} \mu$$

The inverse covariance weights deviations by how surprising they are relative to ordinary co-movement.

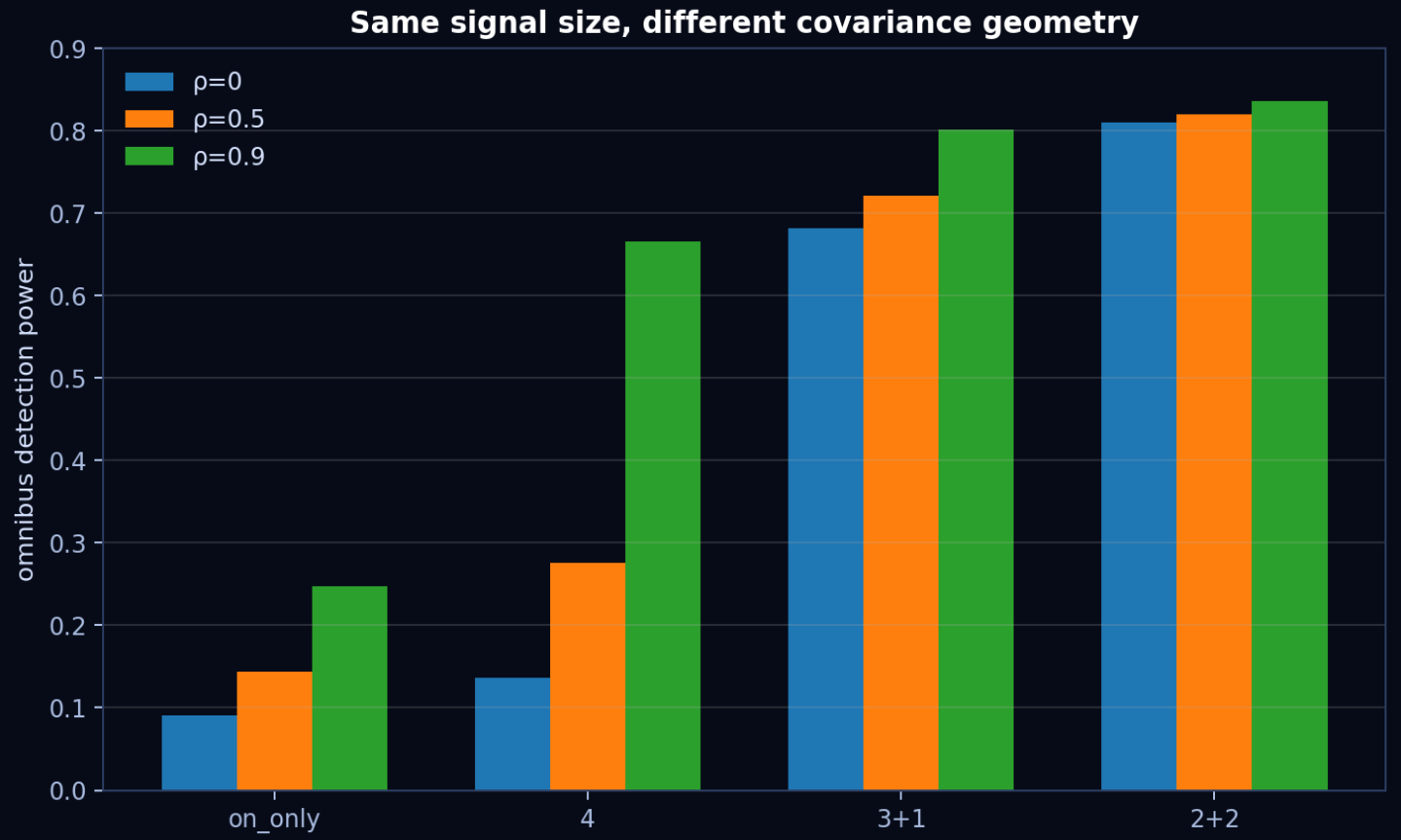
A large raw deviation can be ordinary along a noisy eigen-direction; a smaller deviation can be extraordinary along a tightly pinned direction.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

The same signal size can have radically different power

PROTECTED GEOMETRY

Cluster alignment survives even when Σ is known exactly.



WHY THIS MATTERS

A detector with perfect covariance knowledge is still vulnerable to footprints aligned with intrinsically noisy cluster directions. Estimation error is an additional penalty, not the whole problem.

This is why the protected stage had to be solved first.

Source: protected_attack_geometry_v1_4_2.csv.

The compact summary: concealment capacity

SUMMARY

Normalize a feasible footprint direction u to fixed physical size.

$$C^*(\Sigma, \mathcal{U}) = \max_{u \in \mathcal{U},\ \|u\|=1} \frac{1}{\sqrt{u^T \Sigma^{-1} u}}$$

$$\mathcal{U} = \mathbb{S}^{d-1} \quad \Rightarrow \quad C^* = \sqrt{\lambda_{\max}(\Sigma)}$$

READ IT LIKE THIS

The best hiding direction is the physically feasible direction with the most unexplained background variance. If every direction were possible, it would simply be the noisiest eigen-direction of Σ.

PLAIN ENGLISH

“How noisy is the city?” becomes a precise question: how much variance is left along the directions the activity actually has to occupy after the observer has calibrated the joint background?

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

VERIFICATION

How we verified the protected benchmark

Five invariants attack five different failure modes.

INVARIANT	STATUS	BASIS
Model reduction ρ=0,c=1	PASS	exact matrix identity
Common-mode contrast cancellation	PASS	exact contrast identity
Null size at nominal 1%	PASS	observed |error| = 0.000005
Analytic power vs seeded simulation	PASS	worst = 1.84 Monte Carlo SE
Regression to 675 v1.3.3 cells	PASS	residual SD = 0.9594

Source: protected_invariants_v1_4_2.csv.

Reproducibility caught a bug that “all tests pass” did not

REPRODUCIBILITY

Same seed was not enough when the covariance sampler used a non-unique SVD basis.

BEFORE

rng.multivariate_normal(Σ)

Default SVD factorization can choose different bases inside repeated-eigenvalue subspaces on different LAPACK builds.

AFTER

method='cholesky'

Positive-definite Cholesky factorization is unique. v1.4.2 regenerated all five outputs byte-for-byte on a different NumPy/LAPACK stack.

PLAIN ENGLISH

Reproducibility is not housekeeping. It is an adversarial test of the computation itself.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

What we still do not know

LIMITS

A strong research program is explicit about which uncertainty remains outside the model.

- Whether any real hidden actor follows the utility functions or reach constraints in the synthetic game.
- Whether a real sensor system has a stable covariance structure that is well described by our current family.
- How much finite calibration data T is needed before estimation error becomes negligible.
- How robust full-covariance and structured estimators remain when calibration itself is poisoned.
- Whether operational spread/mixed attacks create new failure modes on real multivariate time series.
- Whether any motivating speculative population exists at all.

PLAIN ENGLISH

The next scientific step is not to make the story more elaborate. It is to expose the framework to real backgrounds that were not generated by us.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

03

How to move from synthetic truth to empirical truth

Use real backgrounds, preserve their correlations, inject controlled footprints, and test whether the framework predicts where detection succeeds and fails.

The real-data protocol

PROTOCOL

We can test the framework without needing a real adversary.



Key design rule

Do not destroy the real covariance by shuffling channels independently. Inject the synthetic footprint into the intact multivariate background so common causes, schedules, and sensor coupling remain real.

PLAIN ENGLISH

This turns a speculative detection problem into a standard controlled experiment: the background is real, the hidden signal is known, and success or failure is measurable.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

What makes a useful dataset?

DATASET CRITERIA

The best benchmark is not the biggest one; it is the one that stresses the assumptions we care about.

Multivariate	Several channels measured together
Temporal	Enough clean history to estimate background
Correlated	Real common causes and schedules
Interpretable	We know what at least some channels physically mean
Ground truth	Known faults/attacks help separate detection from labeling error
Accessible	Reproducible use by other researchers

PLAIN ENGLISH

A perfect dataset would include clean calibration, known intervention periods, and physical constraints on which cross-channel footprint directions are feasible.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

Example dataset: SWaT industrial water treatment

INDUSTRIAL CONTROL

A cyber-physical system where attacks, process physics, and correlated sensors meet.

WHY IT FITS

- Multiple process sensors and actuators
- Strong physical coupling across channels
- Known cyberattack scenarios
- Natural test of common-cause covariance
- Operational meaning for false alarms

EXPERIMENT

1. Fit Σ on attack-free operation.
2. Reconstruct contrast directions among tanks, flows, pressures, pumps, and valves.
3. Ask which documented attacks align with high-variance directions.
4. Compare naive independent-channel alarms with Mahalanobis / mixed-policy alarms.
5. Repeat with finite-T covariance estimation.

Source: Singapore University of Technology and Design iTrust SWaT testbed documentation.

External dataset source: iTrust / SUTD SWaT testbed.

Example dataset: MetroPT-3 predictive maintenance

PREDICTIVE MAINTENANCE

Real railway compressor telemetry gives us an excellent non-adversarial transfer test.

1,516,948	15	0.1 Hz
time-series records	features / sensors	collection frequency

TEST THE CORE CLAIM WITHOUT AN ADVERSARY

Use normal compressor operation as the correlated background. Treat documented fault intervals as naturally occurring footprints. Estimate whether fault signatures appear in low- or high-variance covariance directions, then compare covariance-aware scores with per-sensor thresholding.

Why valuable: if geometry predicts detectability even when nobody is trying to hide, that validates the statistical mechanism independently of the adversarial story.

Source: UCI MetroPT-3, DOI 10.24432/C5VW3R.

Example dataset: NASA SMAP / MSL telemetry

SPACE TELEMETRY

Rare anomalies across spacecraft subsystems test multivariate detection under extreme base-rate pressure.

WHY IT FITS

- Real spacecraft telemetry
- Labeled anomaly intervals tied to actual mission issues
- Many channels with subsystem coupling
- Rare-event setting where false positives matter
- Natural sequential / change-point extension

EXPERIMENT

Estimate a clean covariance model from nominal telemetry, then ask whether documented anomaly directions have low Mahalanobis norm because they align with ordinary spacecraft operating modes. Compare single-channel detectors with joint residual tests and evaluate false-discovery burden.

Source: Hundman et al., KDD 2018 / JPL Telemanom repository.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

Example dataset: UNSW-NB15 network intrusion

CYBERSECURITY

High-dimensional correlated traffic is where “match each feature separately” most obviously fails.

2.54M	49	9
records	derived features	attack families

EXPERIMENT

Treat normal traffic as the background distribution. Measure how attacks depart from the joint feature covariance, then deliberately construct mimicry variants that preserve more marginals while changing cross-feature structure. The question is whether the covariance-aware detector degrades gracefully as the attack becomes more distribution-matched.

This is the closest real-world analogue to our “hider chooses u” model.

Source: UNSW Research, UNSW-NB15 dataset.

PHYSICAL SYSTEMS

Example dataset: hydraulic condition monitoring

A controlled physical test rig gives us labeled degradation with many sensor types and known load cycles.

2,205
60-second cycles

17+
sensor / virtual channels

4
fault families

WHY IT IS USEFUL

The load cycle is repeated, so the observer gets repeated opportunities to estimate covariance under comparable operating conditions. Cooler, valve, pump, and accumulator degradation create distinct footprint directions. That is almost tailor-made for testing whether $u^T\Sigma^{-1}u$ predicts which faults are easy or hard to detect.

Source: UCI Condition monitoring of hydraulic systems, DOI 10.24432/C5CW21.

Example dataset: ToN-IoT heterogeneous telemetry

SENSOR FUSION

This one lets us ask whether a footprint can hide in one modality but fail across modalities.

MODALITIES

- IoT / IIoT sensor telemetry
- Network traffic
- Linux audit / resource activity
- Windows performance traces

FRAMEWORK TEST

Fit separate and joint background models. Ask whether an event that looks ordinary inside one modality becomes improbable when network, host, and sensor channels are fused. Then simulate limited-reach contamination: what happens if the hider can manipulate network traffic but not host counters, or sensors but not audit logs?

This directly operationalizes “multi-modal coincidence only helps when ordinary cross-modal dependence is modeled.”

Source: UNSW Research, ToN-IoT datasets.

APPLICATIONS

Where the framework could matter

The same mathematics appears whenever a rare process can adapt to a detector or hide inside correlated background variation.

Industrial cybersecurity	Attacks that preserve process-consistent sensor patterns
Predictive maintenance	Fault signatures aligned with normal operating modes
Network intrusion	Mimicry and poisoning against learned normal traffic
Spacecraft / aviation	Rare anomalies in strongly coupled telemetry
Fraud / abuse	Adaptive behavior responding to published scoring rules
Scientific instrumentation	Multi-sensor anomalies with common-cause noise

PLAIN ENGLISH

The transfer claim is not “our detector solves all these domains.” The transfer claim is narrower: covariance geometry, reach limits, estimator choice, and adaptation are recurring failure modes worth testing.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

ALGORITHM

A controlled injection experiment

Real background + synthetic footprint lets us measure cause and effect cleanly.

PSEUDOCODE

```
split real background into calibration / test
estimate Σ̂ on calibration
choose normalized feasible footprint u
for amplitude a in grid:
    x'_t = x_t + a·u during injected window
    score x'_t with naive and covariance-aware tests
    measure size, power, delay, false discoveries
repeat over u, T, sensor subsets, and random seeds
```

THE KEY OUTPUT

$$\text{power}(a, u, T) \quad \text{vs.} \quad u^T \Sigma^{-1} u$$

If the framework is right, footprint geometry should explain a meaningful share of which injected signals are easy and hard to detect.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

The next stage: finite-T covariance estimation

V1.4 NEXT

The observer will no longer know Σ; it must learn it from clean calibration periods.

$$\frac{T - k}{k(T - 1)} Q \sim F_{k, T-k}$$

Known contrast mean; covariance estimated from T independent calibration periods.

$$\Delta_T(w) = U^*_{\text{est}}(w, T) - U^*_{\text{protected}}(w) \geq -\epsilon_{\text{MC}}$$

WHY THIS MATTERS

Δ tells us the price of learning the background instead of knowing it. As T grows, the estimated frontier should approach the protected frontier. That curve becomes a practical calibration-data requirement.

PLAIN ENGLISH

This is the first result that can directly tell a real observation program how much clean baseline data it needs before covariance uncertainty stops dominating.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

DEMO

What a live demo should show

One clean visual experiment is better than a wall of equations.

1. Real data

Show 4–8 correlated sensor channels.

2. Inject signal

Add the same-sized footprint in two directions.

3. Compare tests

Naive per-channel threshold vs. covariance-aware score.

4. Reveal geometry

The “smaller” anomaly can be easier because it violates covariance.

PLAIN ENGLISH

The audience does not need to understand matrix inversion to see the point: “big” is not the same as “surprising,” and correlations decide the difference.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

What would falsify or weaken the framework?

PRECOMMITMENT

We should know the failure conditions before touching the real datasets.

- Mahalanobis geometry does not predict relative detectability of controlled footprints better than simple marginal variance.
- Estimated-covariance tests fail nominal size even when finite-sample F calibration is used correctly.
- ΔT(w) does not shrink toward the protected benchmark as clean T grows, beyond predeclared Monte Carlo tolerance.
- Structured estimators appear superior only because cluster assignments were tuned using the test anomalies.
- Randomized sensor/test policies lose their advantage once realistic temporal dependence is preserved.
- Results depend on one dataset, one preprocessing pipeline, or one arbitrary normalization of u.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

READING

Intellectual anchors

Our framework is new in combination, not ex nihilo.

Huber 1964	robust contamination
Cachin 1998	information-theoretic concealment
Rousseeuw & Van Driessen 1999	robust covariance
Axelsson 2000	base-rate problem
Dalvi et al. 2004	adversarial classification
Kloft & Laskov 2010/2012	adversarial anomaly detection
Li & Yu 2021	robust change-point detection
MacKenzie et al. 2002	imperfect detection / occupancy

PLAIN ENGLISH

These fields independently rediscovered pieces of the same structure: imperfect observation, contamination, adaptive response, correlated sensors, and the danger of confusing rarity with evidence.

Source: verified v1.3–v1.4.2 project artifacts unless otherwise noted.

WHAT WE KNOW

Detection is a geometry-and-information problem, not simply an anomaly-counting problem.

1. Background uncertainty can dominate raw anomaly counts.
2. Adaptive actors exploit estimator and control vulnerabilities.
3. Randomization helps when reach and information are limited.
4. Covariance geometry determines which physical directions are naturally concealed.

NEXT: prove the mechanism on real multivariate backgrounds, then study finite-T estimation and poisoned calibration.

And keep the epistemic boundary explicit: none of this is evidence that the original speculative population exists.